

Week 1 Weekend Assignment

"Network Detective Challenge"

Due: End of Week 1 (before Day 5 test) | Time estimate: 3-4 hours across Saturday + Sunday

What you need: Your own laptop/PC, Wireshark (free), internet connection, this document

The Big Picture

Over the past 4 days you learned:

- Day 1: What cybersecurity is — CIA Triad, AAA, threats vs vulnerabilities vs risks
- Day 2: Networking basics — OSI model, TCP/IP, IP addressing, ports, protocols
- Day 3: Common protocols — HTTP/HTTPS, FTP/SFTP, SSH, SMTP, DNS, ICMP + subnetting
- Day 4: Linux & Windows security — permissions, users, hardening, event logs

Now it is time to combine all of that into one real-world mini-project.

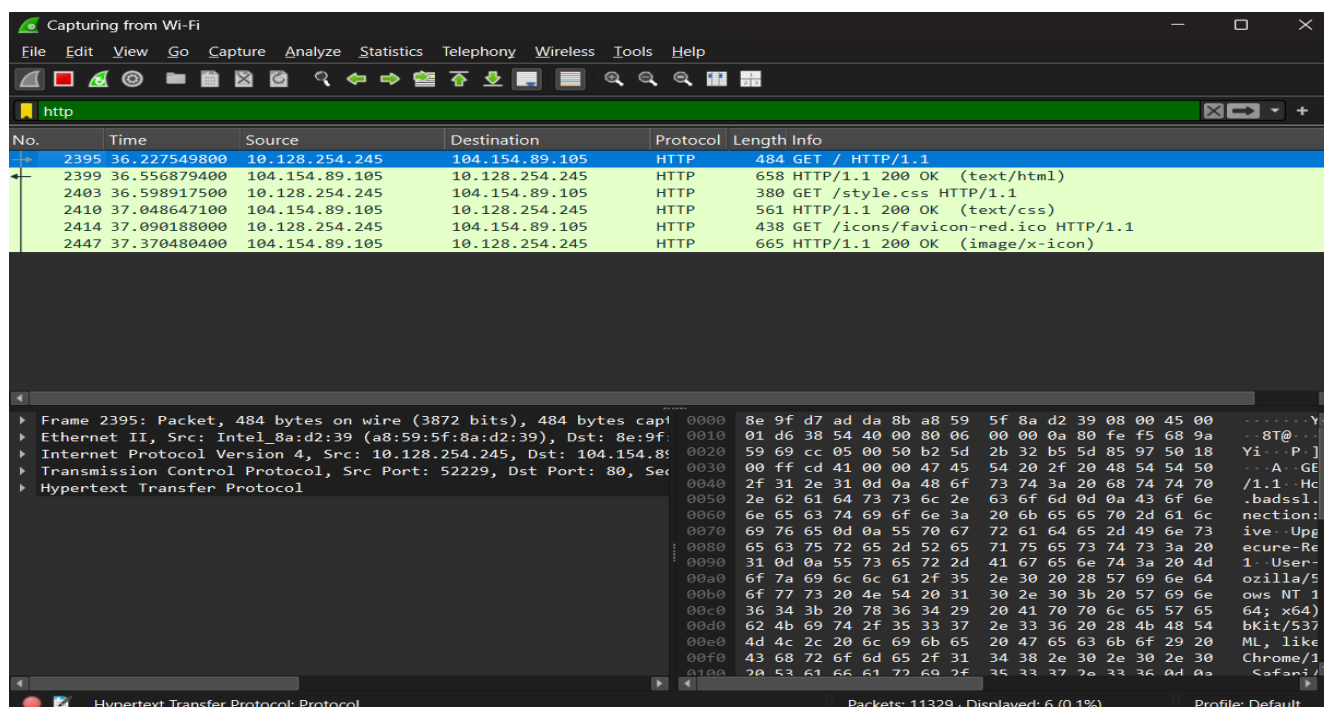
This assignment has 3 parts. Do all 3. They build on each other.

Part 1 — Capture & Analyze HTTP Traffic (Day 2 + Day 3 skills)

What to do:

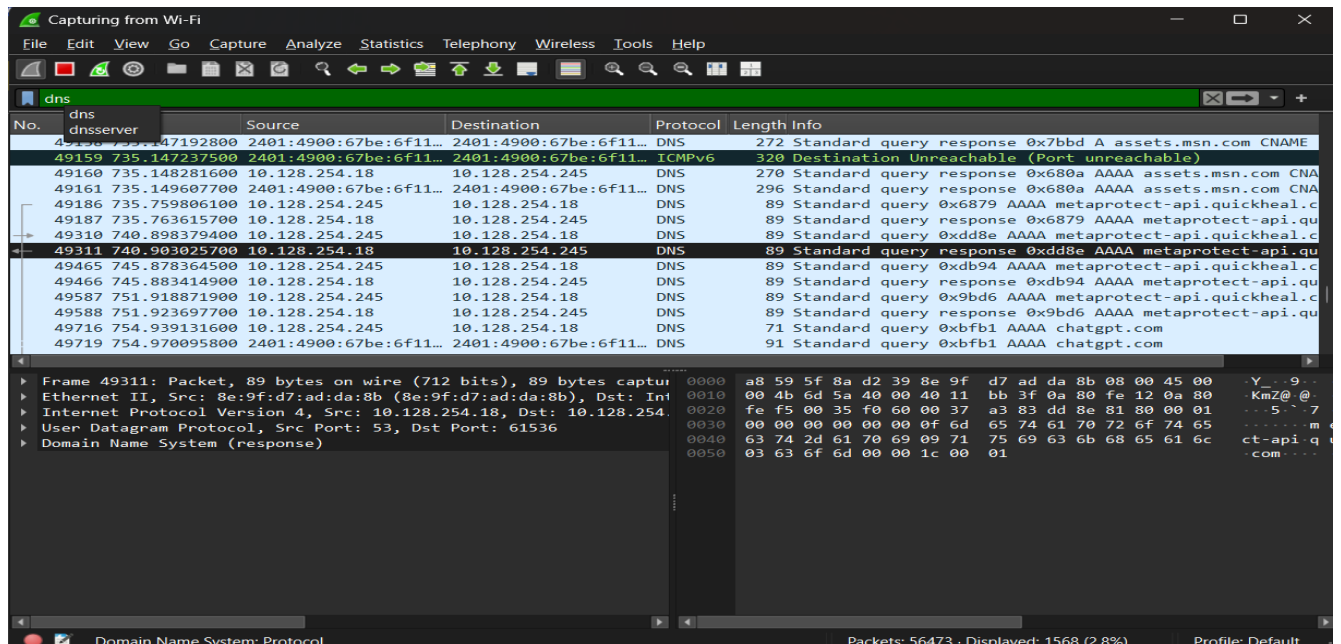
1. Install Wireshark (if not already): <https://www.wireshark.org/download.html>
2. Open Wireshark and start capturing on your active network interface (Wi-Fi or Ethernet).
3. While capturing, open a browser and visit 3 different websites:
 - One that uses HTTP (try <http://neverssl.com> — it deliberately uses HTTP)
 - One that uses HTTPS (try <https://www.google.com>)
 - One of your choice (any site you normally use)
4. Stop the capture after about 60 seconds of browsing.
5. Apply the display filter http in Wireshark — this shows only HTTP traffic.
6. Also try the filter dns to see DNS queries.
7. Also try the filter tcp.port == 443 to see HTTPS traffic.

HTTP Filter in Wirshark:



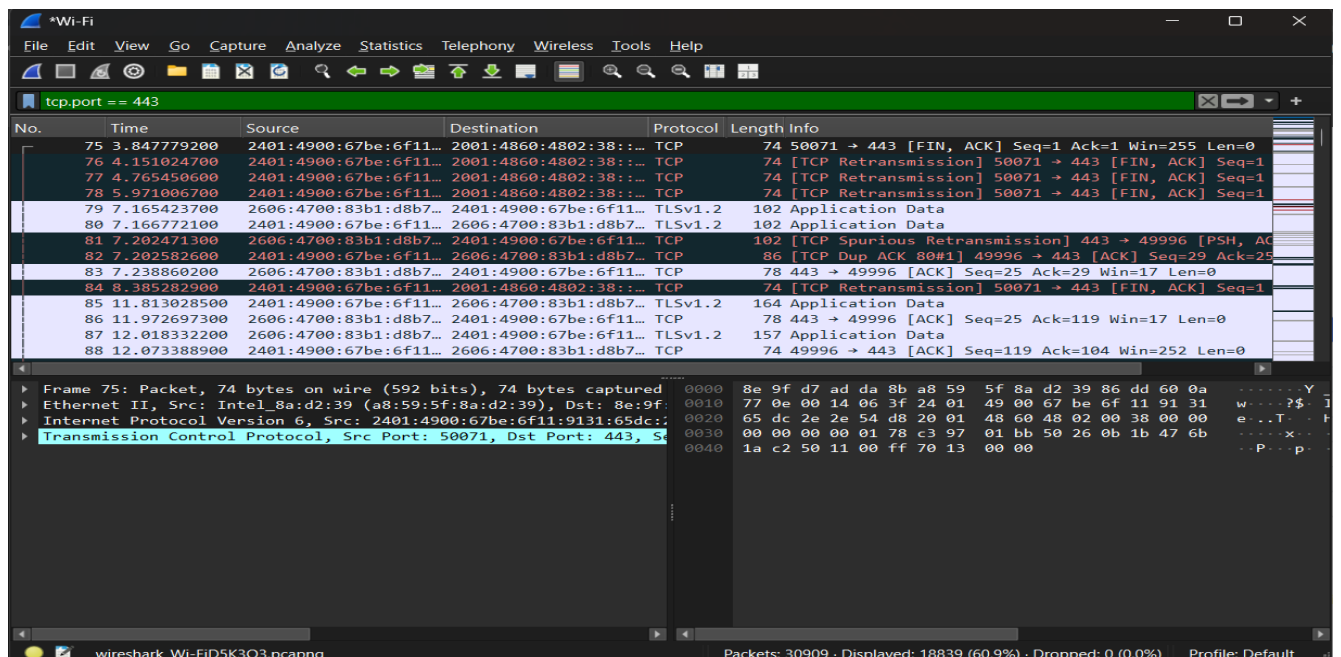
Question	Your answer
What website did you visit?	neverssl.com
HTTP or HTTPS? How can you tell from the capture?	HTTP
What IP address did your computer connect to?	Source Address: 10.128.254.245 Destination Address: 34.218.221.118
What was the destination port? (80 or 443?)	80
For HTTP sites: what information is visible in the packet?	website name, URL, HTTP GET request
For HTTPS sites: what information is visible vs hidden?	-
What DNS queries did you see before the connection?	Google.com

DNS filter in wireshark

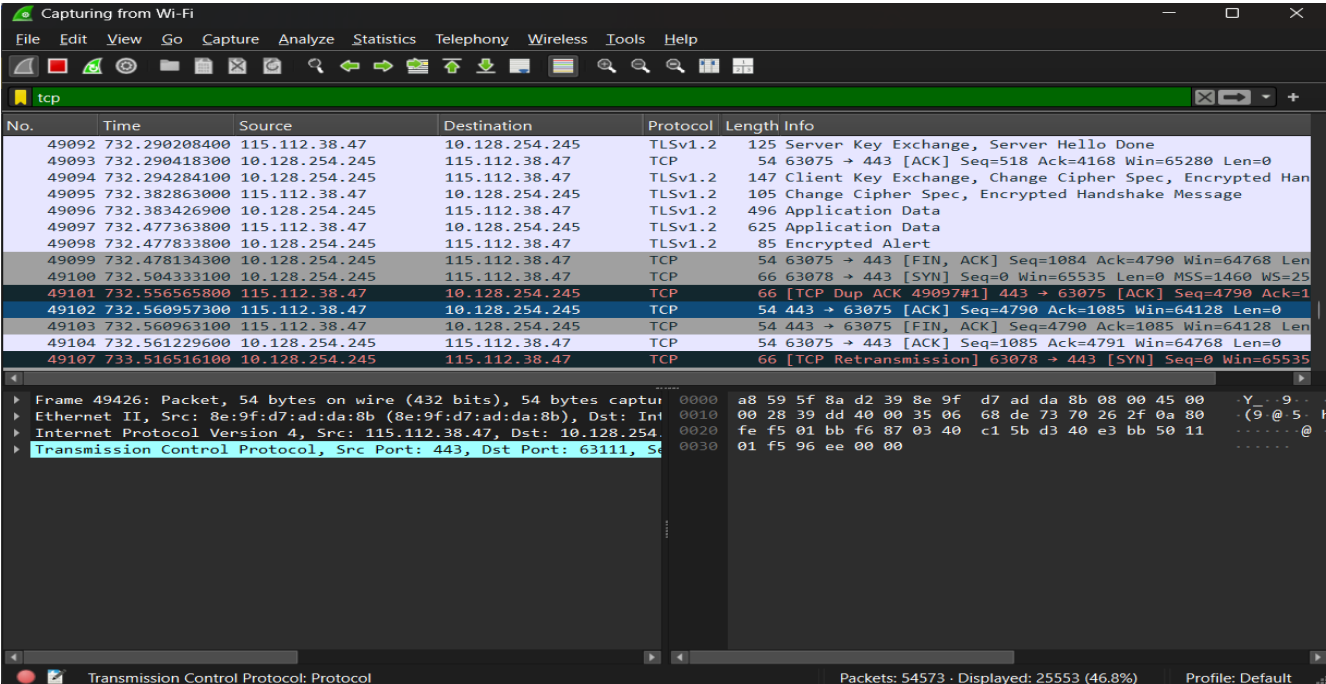


I open google.com, the domain is google.com

TCP PORT 443



HTTPS filter in Wireshark



Question	Your answer
What website did you visit?	youtube.com
HTTP or HTTPS? How can you tell from the capture?	HTTPS
What IP address did your computer connect to?	Destination port 443
What was the destination port? (80 or 443?)	Source Address: 2401:4900:67be:6f11:9131:65dc:2e2e:54d8 Destination address:2001:4860:38::178
For HTTP sites: what information is visible in the packet?	IP addresses
For HTTPS sites: what information is visible vs hidden?	Data is encrypted
What DNS queries did you see before the connection?	Google.com

Key concepts to connect:

- CIA Triad: When you visited the HTTP site, which part of CIA was violated? (Hint: no encryption = no Confidentiality)
- AAA: If the HTTP site had a login page, which AAA function would be exposed? (Hint: Authentication credentials)

in clear text)

- Subnetting: What is the IP range of your home network? (Check with ipconfig or ifconfig) What subnet mask are you using?

Where to get help / sources:

- Wireshark official docs: <https://www.wireshark.org/docs/>
- Wireshark 101 tutorial: <https://www.youtube.com/watch?v=TkCSr30UojM>
- How to filter HTTP traffic: In Wireshark, type http in the display filter bar

Part 2 — Subnetting Practice (Day 3 skills)

Using what you learned about subnetting in Day 3, solve these problems. Show your working.

Problem 1

Your home router gives you the network 192.168.1.0/24.

- How many total host addresses are available in this network? 256
- If you split it into 4 equal subnets, what is the new subnet mask in CIDR notation? 26
- What are the network addresses of all 4 subnets?
 - 192.168.1.0/26
 - 192.168.1.64/26
 - 192.168.1.128/26
 - 192.168.1.192/26
- What is the broadcast address of the third subnet?
 - 192.168.1.128/26
- How many usable hosts per subnet after splitting?
 - $2^{(32-26)} - 2 = 2^6 - 2 = 62$

Problem 2

You are a network admin for a small office. You have these departments:

- Sales: needs 30 hosts
- Engineering: needs 50 hosts
- HR: needs 10 hosts
- Servers: needs 10 hosts

You are given the network 10.0.0.0/24.

- Design a subnetting scheme that gives each department enough addresses.
- For each department, write: network address, subnet mask (CIDR), usable host range, broadcast address.
- Which department gets the largest subnet? Why?

Department	Network Address	CIDR	Usable Host Range	Broadcast Address
Engineering	10.0.0.0	/26	10.0.0.1 - 10.0.0.62	10.0.0.63
Sales	10.0.0.64	/27	10.0.0.65 - 10.0.0.94	10.0.0.95
HR	10.0.0.96	/28	10.0.0.97 - 10.0.0.110	10.0.0.111
Servers	10.0.0.112	/28	10.0.0.113 - 10.0.0.126	10.0.0.127

Problem 3 — Connect to Security

- Why would a security team want to put the Servers department on a different subnet from Sales? (Answer using CIA + AAA concepts)

Separating servers into a different subnet improves **Confidentiality, Integrity, and Availability (CIA)** by protecting important data. **AAA** controls who can access the servers and tracks user activities.

- If an attacker compromises a Sales PC, can they directly access a Server on a different subnet? Explain why or why not.

No, because traffic between subnets is controlled by routers and firewalls. The attacker would need proper permissions or exploit a vulnerability to access the server.

- What device controls traffic between subnets? (Hint: think about Day 2 — routers/firewalls)

A **router** or **firewall** controls traffic between subnets and enforces security rules.

Part 3 — Connect Everything: CIA + AAA Across All 4 Days

This is the thinking part. Write short paragraphs (3-5 sentences each). No copy-paste — use your own words.

Question 1 — CIA Triad in Action

Pick one of the protocols you studied (HTTP, FTP, SMTP, DNS, SSH, or ICMP). For that protocol:

- How does it support one part of the CIA Triad?
- How does it fail to protect one part of the CIA Triad?
- What would a security professional do to fix that weakness?

I chose HTTP as the protocol. HTTP helps with **Availability** because it allows users to access websites easily. However, it does not protect **Confidentiality** because data is sent without encryption. A security professional can fix this by using **HTTPS**, which encrypts the data and keeps it secure.

Question 2 — AAA Framework in Action

Think about logging into a website (like Gmail or any site you use).

- Authentication: How does the site verify you are who you say you are?
A website checks my identity using a username, password, and sometimes OTP or multi-factor authentication.
- Authorization: After you log in, what are you allowed to do? What are you NOT allowed to do?
After logging in, I can access my own account and data, but I cannot access other users' accounts.
- Accounting: What records might the site keep about your login? Why?

The website keeps records such as login time, IP address, and user activities for security and monitoring purposes.

Now answer: What happens if an attacker captures your HTTP traffic from Part 1? Which part of AAA breaks first? Why?

If an attacker captures HTTP traffic, **Authentication** breaks first because usernames and passwords can be seen in plain text. The attacker can then use those credentials to log in as the user.

Question 3 — The Full Attack Story

Imagine this scenario:

An employee at a company uses an unencrypted HTTP connection to log into the internal HR portal from a coffee shop. The company's network is poorly segmented — all departments are on the same subnet 192.168.1.0/24.

Using everything you learned this week, write a short attack story:

1. What can the attacker on the coffee shop Wi-Fi see? (Day 3 — HTTP, packet capture)

The attacker can capture network traffic and see the employee's username and password because HTTP is not encrypted.

2. What can the attacker do with those credentials? (Day 1 — Authentication broken, CIA violated)

The attacker can log in to the HR portal as the employee and access sensitive information.

3. Once inside the company network, what can the attacker reach? (Day 3 — subnetting, no segmentation)

Since all departments are on the same subnet, the attacker may be able to access other systems and shared

resources on the network.

4. How could the company have prevented this? (Day 4 — hardening, Day 3 — segmentation, Day 1 — controls)
The company could use HTTPS, enable multi-factor authentication, and separate departments into different subnets with firewall protection.

Question 4 — Linux/Windows Connection

- On Day 4 you learned about file permissions (Linux) and Event Viewer (Windows).
- How does the principle of least privilege (Day 4) connect to the AAA Authorization function (Day 1)?
Least privilege means users only get the access they need. This supports Authorization by limiting unnecessary permissions.
- If an attacker gets a shell on a Linux machine, what permission-related files would they look for?
They may look at files such as sudo configuration files to find user and permission details.
- On Windows, where would you look in Event Viewer to spot a brute-force login attempt?
I would check **Windows Logs** → **Security** for repeated failed login attempts and suspicious login activity.

How to Submit

Create a single document (text file, markdown, or PDF) with:

1. Part 1: Your traffic analysis table + answers + at least 2 screenshots from Wireshark
2. Part 2: All subnetting answers with working shown
3. Part 3: All 4 written answers in your own words

Name it: Week1_Weekend_Assignment_YourName.md (or .txt or .pdf)

Grading Rubric (self-grade before submitting)

Criteria	Points	Your score
Part 1: Traffic captured and analyzed correctly	/25	
Part 1: CIA + AAA connections made	/10	
Part 2: Subnetting calculations correct	/25	
Part 2: Security reasoning about subnets	/10	
Part 3: Own words, no copy-paste	/15	
Part 3: Connections across all 4 days are accurate	/15	
Total	/100	

Tips for Doing This Well

- Do Part 1 first. It is the most hands-on and will make the theory in Parts 2 and 3 feel real.
- Use your notes from all 4 days. This assignment is designed to make you go back and review everything.

- If you get stuck on subnetting, re-read the Day 3 notes. The 192.168.1.0/24 example is your starting point.
- Screenshots matter. In cybersecurity, evidence matters. Show what you captured.
- Write like you are explaining to a friend. If you cannot explain it simply, you do not understand it yet.